

EXERCISE 1

The company TECH-INOV, operating in a monopoly situation, produces computers using the following cost function:

$$C(Q) = 50 + 40Q$$

The direct demand function addressed to TECH-INOV is given by:

$$Q = 32 - 0.4P$$

with Q being the quantity of computers produced and P the selling price of a computer.

Answer the following questions:

1. What do you mean by “monopoly”?
2. Determine the optimal quantity of computers to produce, the selling price of a computer, and the value of the profit achieved.
3. Define, determine, and interpret the price elasticity of demand for computers.
4. As a consumer, on what type of market would you prefer to purchase the computer? Clearly justify your answer.
5. Present two problems arising from monopoly pricing.

EXERCISE 2

The company GITECH wishes to engage in the production and sale of computer software. This project will have a lifespan of 10 years. The total amount of the project is equal to 18 million CFA francs.

- Years 1–7: Cash flow = 4 million CFA francs per year
- Year 8: Expected revenue = 2.5 million CFA francs
- Year 9: Expected revenue = 6 million CFA francs
- Year 10: Expected revenue = 1 million CFA francs

The discount rate chosen by the company is linked to the average profitability of funds invested by GITECH over the past 15 years. This rate is 10%.

Answer the following questions :

1. Show how to evaluate the profitability of an investment through the Net Present Value (NPV). (1 pt)
2. In what case should NPV be used to compare two or more different projects? (1 pt)
3. Should GITECH engage in this project? Clearly justify your answer. (2 pts)

EXERCISE 3

The mobile telephony market in country Alpha includes one company in a dominant position and ten identical small companies. This market evolves in a competitive context.

- Total market demand: Q
- Demand function: $P = 200 - Q$
- Dominant company marginal cost: 40 monetary units
- Small company production: q_p with marginal cost $c_p = 40 + 10q_p$

Answer the following questions:

1. Is the good referred to in the mobile telephony market of country Alpha of zero marginal cost? Clearly justify your answer.
2. Is a leading company on a market necessarily in a dominant position? Justify your answer with a clear example.
3. Formulate the supply function of all the small companies as well as the demand function addressed to the dominant company.
4. Determine the market equilibrium as well as the market share of each company. Comment on your results.

EXERCISE 4

The company ANDY is engaged in the production and sale of a computer game.

- Fixed costs: 800 million CFA francs

- Selling price: 400 CFA francs per copy
- Tax: 20% per copy sold

Answer the following questions:

1. Considering that the game has zero marginal cost, how many copies must be sold to cover the fixed costs?
2. How many copies must be sold to cover the fixed costs if the marginal cost is 120 CFA francs?
3. Under what condition is the marginal cost of this computer game zero?
4. Under what condition is the marginal cost positive?

EXERCISE 5

The company ANDY wishes to engage in the production and sale of computer software. This project will have a lifespan of 40 years. The total amount of the project is equal to 18 million CFA francs.

- Years 1–10: Cash flow = 4 million CFA francs per year
- Year 11: Expected profit = 8 million CFA francs
- Years 12–40: Expected annual revenue = 6 million CFA francs

The discount rate chosen by the company is linked to the average profitability of funds invested by ANDY over the past 20 years. This rate is 10%.

Answer the following questions:

1. Should ANDY engage in this project? Clearly justify your answer.
2. Determine and comment on the profitability index. Comment on your result.